

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. `<media src="video.mp4"></media>`

- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.

A. Correct Option: B.

`<video src="video.mp4" controls></video>`

Justification:

- The `<video>` tag is the standard HTML5 element for embedding videos.
- The `controls` attribute provides play, pause, volume, and fullscreen controls.
- Other options (`<media>`, `<movie>`, `<vid>`) are **not valid HTML5 tags**.

2. Identify the **appropriate HTTP method for form submission** and explain why.

A.

Correct Option: B. POST

Explanation:

- The **POST** method sends user data in the **HTTP request body**.
- It is suitable for registration forms because sensitive information is **not displayed in the URL**.
- POST is more secure than GET for submitting user information.

3. Choose the **proper semantic element for navigation** and justify its usage.

A.

Correct Option: B. <nav>

Explanation:

- The **<nav>** element defines a section containing navigation links.
- It improves HTML semantics, accessibility, and search engine optimization (SEO).
- Screen readers can easily identify navigation menus.

Example:

```
<nav>

  <a href="home.html">Home</a>

  <a href="about.html">About</a>

  <a href="contact.html">Contact</a>

</nav>
```

4. Identify the **correct XML syntax** and explain the error in other options.

A.

Correct Option: B.

<note>

<to>User</to>

<from>Admin</from>

</note>

Errors in Other Options:

A. <note><to>User</to><from>Admin</from>

- Missing closing </note> tag.

C. <note><to>User</from></note>

- Mismatched tags (<to> is closed with </from>).

D. <note to="User"></note

- Missing closing > in the end tag (</note>).

5. Choose the **correct protocol for web communication** and justify your choice.

A.

Correct Option: B. HTTP

Explanation:

- HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers.
- It is used to transfer HTML pages, images, videos, CSS, JavaScript, and other web resources.

Other Protocols:

- FTP – Used for file transfer.
- SMTP – Used for sending emails.

- SNMP – Used for network management and monitoring.

Code of Conduct:

I Abburi Ganesh Ram Kamal (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable	Limited clarity and structure	Poor clarity; difficult to understand

			presentation		
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